Mapping Urban Development And Gentrification In Dallas Texas

Final Research Paper:

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EPPS 6302 Methods of Data Collection and Production

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Introduction

As Dallas emerges as a hotspot for economic investment, it has become a prime target for urban redevelopment and real estate speculation (Snyder 2024). These rising investments have contributed to the displacement of vulnerable homeowners, particularly in historically Black and Latino neighborhoods. Gentrification, which is a process in which lower-income urban areas are transformed by an influx of wealthier residents and capital, has accelerated across the city through targeted investments in housing and commercial infrastructure (Rajwani-Dharsi 2024). One in five Dallas neighborhoods are currently in the dynamic or late stages of gentrification, marked by sustained housing market growth and demographic turnover (Snyder 2024). The same study found that average home sale prices in these areas rose from approximately $55,000 in 2011–2012 to $238,000 by 2021–2022, reflecting a dramatic shift in affordability and access. These changes disproportionately affect Latino and Black communities, who face heightened risk of displacement and cultural erasure. Consequently, our project seeks to provide the necessary twenty-five-year quantitative examination to establish the precise correlation between urban investment, escalating housing unaffordability, and the resultant racial and ethnic demographic transformation across Dallas.

Literature Review

Our project operates under the foundational premise that data is produced, not found, a concept supported by Matthew J. Salganik that highlights the nature of modern datasets. In the digital age, much of the information available for studying urban change, such as administrative records or digital traces, is readymade as it is data created by governments or companies for purposes like billing or service provision other than scientific research (Salganik 2019). Establishing this data production nature requires the application of methods, such as the hybrid strategy, that consciously blend large-scale readymade data, such as those available via United States (U.S.) Census and Federal Reserve Economic Data (FRED) APIs, with customized research inputs that are created to ensure data adequately fits our sociological questions about gentrification and urban development.

Theoretically, our project also aligns with the Algorithmic Modeling Culture championed by Leo Breiman. Rejecting the assumption that data springs from simple, known statistical model as often assumed by the traditional Data Modeling Culture (Breiman 2001). Viewing the data generating process in fields dealing with complex phenomenal, such as urban studies as a "black box" that is "complex mysterious, and at partly unknowable" (Breiman 2001, 209). Therefore, the produced complexity of urban data dictates adopting algorithmic approaches that focus on building predictive models to extract reliable information about the underlying mechanism, rather than fitting data to overly simplistic models that may lead to questionable conclusions due to weak predictive accuracy (Breiman 2001).

The nature of the data generation process to urban studies is complex and unknowable, which makes the commitment to Breiman's Algorithmic Modeling Culture critical to understand gentrification. Many sociological texts confirms that gentrification phenomenon is a multidimensional cultural practice rooted in the dynamics of economic restructuring and shifts in production and consumption, rather than individual decisions (Zukin 1987). By adopting Breiman's algorithmic information extraction approach about the underlying mechanisms of gentrification. Although, Breiman suggests moving away from relying on exclusively on traditional data model that are generated by simple stochastic models or linear regressions (Breiman 2001). We will incorporate a simple Ordinary Least Squares (OLS) regression to serve as the necessary step within a broader algorithmic approach. OLS falls under the Data Modeling Culture, that is designed to estimate parameter based on linear regressions (Breiman 2001). Fitting data under this models could lead to weak predictive accuracy on complex issues. Even if the need is to use complex algorithmic modeling, OLS is a necessary integration for two main reasons: (1) establishing baseline relations between median housing costs, median household income, and owner-occupied share, (2) to evaluate predictive accuracy to serve as a baseline comparison against more sophisticated techniques like Random Forests. If a traditional model obtains better predictive accuracy than simple baseline it would draw questionable conclusions on its information extraction capabilities (Breiman 2001). On the other hand, if a more advanced algorithmic model offers a better predictive accuracy, it would mean it is providing more reliable information about the structure of the relationship (Breiman 2001). Thus, using OLS in our project provides a fundamental diagnostic tool to clarify linear trends and establish a minimum predictive accuracy baseline before progressing to a more advanced, black-box algorithmic methods.

By acknowledging the data production nature, our project aims to commit to a rigorous methodology that can handle complex inputs and yielding more reliable insights into how the City of Dallas gentrification unfolds. The initial research on gentrification focuses heavily on documenting it as a process of neighborhood change and demographic shifts, contemporary urban sociology that emphasizes the dynamic of economic restructuring (Zukin 1987). Analyzing long-term shifts provides a better understanding of housing costs alongside demographic transformation of capital reinvestment and the changing patterns of social differentiation and consumption, rather than attributing it to individual-level choice or demographic structure (Zukin 1987). Our project aims to investigate Dallas's gentrification reflects this complex interplay between reinvestment, demographics, and housing costs.

Problem Statement

The currently literature and policy discussion concerning gentrification in the City of Dallas are missing a long-term quantitative analysis correlating housing increases with racial and ethnic demographic shifts. Although, there has been recent efforts, such as the Builder of Hope report, to track neighborhood change over a significant 10-year period (2011-2021) and document price acceleration and displacement risks for Black and Latino residents. They do not provide the necessary two-decade historical perspective dating back to the 2000s. Therefore, a dedicated examination using historical U.S. Census and American Community Survey (ACS) data is required to establish the relationship between median housing costs and demographic change across an extended two-decade timeframe within the explicit geographic boundaries of the City of Dallas.

Research Question:

Is there a relationship between the increase of median housing costs within the City of Dallas and the resulting changes in median household income, owner-occupied share, and the spatial differentiation of racial and ethnic demographics over the last two decades?

Data Collection & Generation:

To answer this question requires bridging a gap in the literature regarding a long-term, quantitative examination of this relationship within the Census Tracts of the City of Dallas, where more than 40% of neighborhoods are susceptible to or are currently experiencing gentrification. Our quantitative collection will be designated to 20-year data period (2000s, 2010s, 2020s). Further, the technical foundation of this research relies on designing and building a reproducible R/Quarto, utilizing API-assisted methods and web scraping.

Data Acquisition and Harmonization:

1. **Demographic Data:** R tidycensus package to collect and harmonize historical demographic data from the U.S. Census Bureau/ACS APIs).
2. **Historical Housing Costs (Macro-Level):** The R fredr packages will pull housing costs metrics from the Federal Reserve Bank of Dallas APIs, specifically retrieving time-series data for the Dallas-Plano-Irving, TX Metropolitan Statistical Area (MSA). This MSA-level data will require careful filtering or disaggregation in R to isolate statistics applicable to the tracts or boundaries of the City of Dallas.
3. **Recent Real State Metrics (Micro-level):** To look at the change in the Real Estate Metrics from the last five years since the 2020 Census will be sourced from Zillow. Our team will acquire the data via Zillow's public Home Value Index and their archives in order to ensure construct validity and a transparent process.

Analysis and Visualization:

Finally, our team will finalized the dataset, structured primarily at the Census Tract Level will be analyzed using the R ggplot, sf , tigris, tidyverse packages. The analysis will focus on regression and correlation to identify the relationships between demographic shifts and rising median housing costs. Results will be visualized through mapping and charts to highlight the heterogeneity in impacts, focusing on vulnerable neighborhoods such as West Dallas, South Dallas, and Vickery Meadow.

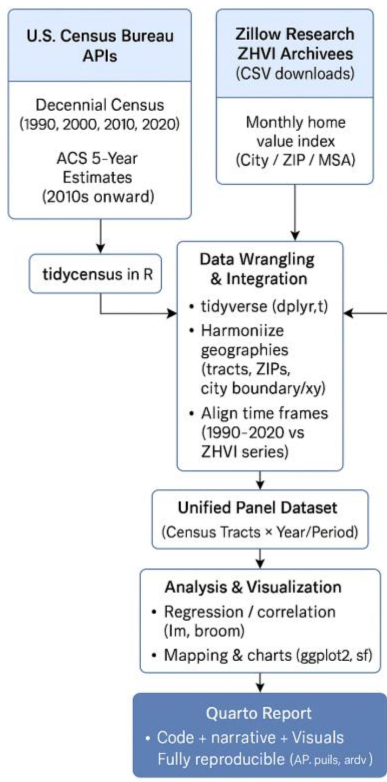


Figure 1 – Data Collection, Generation & Analysis Plan for this project

Data Production & Analysis

Combining datasets is the process of merging information from different sources into one comprehensive dataset. **U.S. Census Bureau (ACS/Decennial Census):** gives demographic details — race, ethnicity, age, education, etc.

- **Zillow:** provides housing metrics such as median listing or sale prices, rent estimates, or price index changes.
- **Federal Reserve Bank of Dallas (FRED):** gives time-series data on median home values and economic indicators.

This **record linkage** process allows us to merge demographic and housing datasets across different time periods (2000–2020) for Dallas County. After linkage, we will clean and standardize all variables to ensure uniform formats (e.g., converting inflation-adjusted housing prices, harmonizing racial/ethnic categories). These linked and labeled datasets will then be used to create **change variables** (e.g., percentage change in home values, demographic shifts) and prepare an analysis-ready dataset for visualization and statistical modeling.

III. Data Generation

Data Acquisition & Production

The data collection process was divided into two parts. The first was collecting demographic data to show shifts in demographics over the years. The second was collecting the house price index and the housing trend in Dallas. To cover the demographic data, the U.S. Census Bureau was used through the `tidycensus` package in R. To capture and understand trends, data were collected for 30 years (2000, 2010, and 2020). Three datasets from the Centennial Census and datasets from ACS 5-year estimates cover gaps between Centennial Census. From the Census variables such as total population, housing units, race and geoid. For the second part of the collection, housing prices were collected using datasets from the Zillow website and `FREDr`, an R package. Variables such as education, income, employment, and housing affordability and price were taken into consideration for analysis.

The cleaning process was fairly simple, as most datasets came from government databases. These datasets are usually cleaned and formatted properly. This was the case for this project; the only data cleaning was trimming, lowercasing, space deletion, and standardization (to merge datasets properly) using `tidyverse` tools such as `dlypr`. Janitor was also used for cleaning and joins to merge all datasets using the geoid column as the primary key. There were additional columns created to prepare for analysis, such as non-Hispanic and aggregated columns, to make a distinction between the demographics observed on this project. This pipeline produced a complete dataset combining demographic, socioeconomic, and housing indicators across 30 years of neighborhood change in Dallas.

Ethics and Compliance

The integrity and transparency of our research paper was maintained through strict adherence to ethical data usage and the principles of reproducibility through following Compliance, Public Data Ethics, Provenance, and Auditability. Our data acquisition was designed to exclusively utilize publicly available, aggregate data from government agencies (e.g. U.S. Census, FRED) and established research indexes (Zillow ZHVI). This approach ensures that we did not collect or analyze and Protected Health Information (PHI) or personally identifiable data. Furthermore, the data analysts ensure safeguards for the R code to comply with Terms of Service for all APIs used. Finally, our project adheres to the principle of transparent provenance, meaning the entire data generating process is fully traceable and auditable. Each of the team members have documentation on all methodological choices, including filtering of the data, handling missing values, and providing any transcripts on AI-usage, thereby ensuring full transparency for the final dataset, presentation, and report. The documentation can be found in the coordinator's Quarto that allows external reviewers to replicate the exact steps of data acquisition and harmonization.

Harmonization and Record Linkage

The construction of a tract-level analytic file for Dallas County (FIPS 48113) required record linkage across three data sources: (1) decennial Census/ACS tract data for 2000, 2010, and 2020; (2) Zillow Home Value Index (ZHVI) data at the ZIP code level for 2000 and 2020; and (3) the HUD ZIP–TRACT residential crosswalk. This approach to integrate the data, shows the foundational premise that data is produced, not found, and utilizes a hybrid strategy blending large-scale, readymade government record with customized research inputs.

First, we cleaned and standardized the Census tract file. The process involved converting the tract GEOID to a character string, removed any trailing decimal artifacts (e.g., ".0"), and left-padded the GEOID to 11 digits. Dealing with these artifacts and ensuring standardization is necessary to handle potentially dirty data and improve overall construct validity. We then filter the data to tracts in Dallas County by keeping only observations whose GEOID begins with the county FIPS code "48113". Using these data, we created a tract-level panel for 2000 and 2020 and reshaped it to wide format so that each tract has values for 2000 and 2020 on the same row.

Second, we prepared the Zillow ZHVI file at the ZIP level. We cleaned variable names, converted ZIP codes to 5-digit character strings, and selected annual mean ZHVI for 2000 and 2020. We reshaped the data so that each ZIP has ZHVI in both years and then computed the change in log ZHVI between 2000 and 2020. We employed the log transformation to ensure home value pricing changes behave like percentage change rather than a dollar change, which provides a stable measure of long-term appreciation for the main explanatory variable (Δ log (ZHVI) = log ($ZHVI_{2020}$) – log($ZHVI_{2000}$)). This transformation helps prepare the data produced for robust statistical analysis, aligning with the goal of extracting reliable information based on the Algorithmic Modeling Culture.

Third, we completed the critical step of record linkage by processing the HUD ZIP–TRACT residential crosswalk, a necessary phase in the data production pipeline. We ensure the standardization required for this complex process by consistently converting the tract and ZIP identifiers (removing decimals and padding to 11-digit GEOID and 5-digit ZIP codes) and restricting the file to tracts in Dallas County. For each unique tract–ZIP pair, we retained the residential ratio (res_ratio), which quantifies the share of residential addresses in the tract that fall within the specified ZIP code.

We then aggregated the prepared ZIP-level Zillow Home Value Index (ZHVI) data to the tract level using the HUD crosswalk, reflecting our hybrid strategy of consciously blending the readymade housing metrics with our research objective. For each tract, the crosswalk was merged with the ZHVI change file by ZIP code, allowing us to compute the residentially weighted mean of the change in log ZHVI (Δ log (ZHVI)) using res_ratio as the official weight. We likewise calculated the weighted mean values of ZHVI for 2000 and 2020 at the tract level.

Finally, we merged all independent data streams to produce the analysis-ready dataset, completing the overall data production pipeline. This involved merging the tract-level demographic change file (2000–2020 Census/ACS data, including measures such as Δ Hispanic), the 2010 baseline controls (such as Log Income and Owner Share), and the newly produced tract-level ZHVI measures by GEOID. This final dataset, structured with one observation per Census tract, contained all variables required for the regression models. This completes the intricate record linkage and aggregation process to build a comprehensive view of urban change in Dallas County, where patterns suggest substantial neighborhood heterogeneity and complex demographic trajectory.

Variable Transformation (The Produced Data)

The following variables were constructed to measure the core outcomes and the long-term change in local housing values, required for the subsequent regression models.

1. Change in Hispanic Share, 2000–2020 (*d_hispanic_pct*)

The primary outcome measures the change in the Hispanic share of each tract's population changed between 2000 and 2020. The variable is calculated by taking the 2020 value and subtracting the 2000 value:

$$\Delta Hispanic = Hispanic_{2020} - Hispanic_{2000}$$

2. Change in Non-Hispanic share, 2000–2020 (*d_non_hispanic_pct*)

We, likewise, calculated for the change in the Non-Hispanic share for each tract's population as a parallel outcome measure, reflecting population shifts over the two-decade period. It is calculated by subtracting the 2020 value minus the 2000 value:

$$\Delta \text{Non-Hispanic} = \text{Non-Hispanic}_{2020} - \text{Non-Hispanic}_{2000}$$

3. Change in Log ZHVI, 2000–2020 (*d_log_zhvi*)

The main explanatory variable captures long-term change in local housing values. To prepare this data for the analysis and ensure that the price changes behave like percent changes rather than dollar changes, a transformation is necessary for reliably extracting information about the underlying mechanism of urban change that is often has mysterious complexity in the Algorithmic Modeling culture, we used the log transformation. For each ZIP code, we computed the change in the natural *log* of the Zillow Home Value Index (*ZHVI*):

$$\Delta \log(\text{ZHVI}) = \log(\text{ZHVI}_{2020}) - \log(\text{ZHVI}_{2000})$$

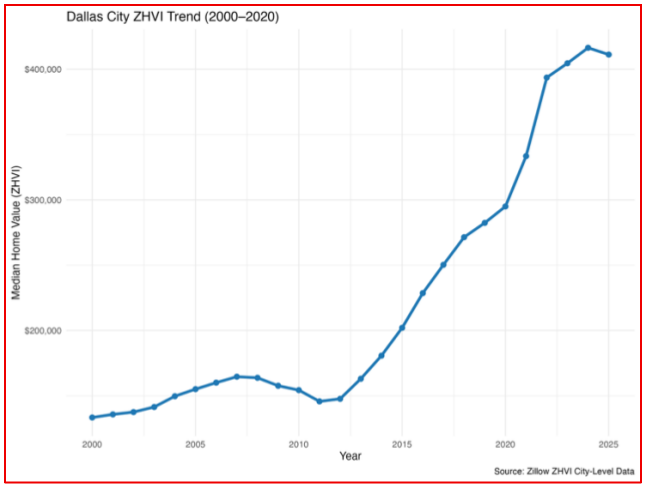
We then aggregated these ZIP-level values to the Census tract level using the HUD ZIP–TRACT residential crosswalk. This required computing a residentially weighted mean of the change in $\Delta \log(\text{ZHVI})$, using the residential ratio (*res_ratio*) as the weight. This final aggregation step produced a tract-level measure of long-term change in housing values from 2000 to 2020.

IV. Analysis and Results

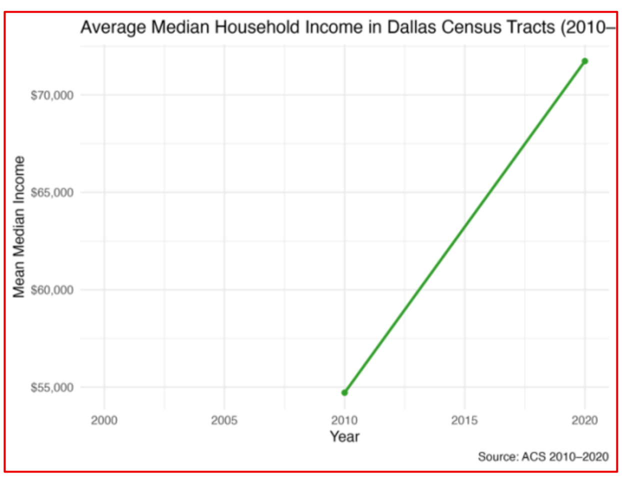
Prior to the estimation of our primary regression modes, we first analyzed the descriptive statistics of our core produced variables. Across Dallas County tracts ($N \approx 821$), the calculated change in Hispanic share (Δ Hispanic) from 2000-2020 exhibited a wide variation, ranging approximately –42 percentage points to +49 percentage points, with a mean increase of +13.9 points. This is a substantial variance in demographic trajectories suggests the operation of complex underlying social mechanisms. Furthermore, the primary explanatory variable, the log change in Zillow Home Value Index (ZHVI) (Δ log (ZHVI)), averaged rough 0.88 when aggregated to the tract level using HUD ZIP-TRACT weights, conforming strong long-run appreciation in real housing values across most neighborhoods.

Baseline socioeconomic conditions in 2010 also displayed notable variability. Specifically, median household income (logged) ranged from approximately ~ 9.25 to ~ 12.43, while the owner-occupied housing share ranged from near zero to nearly 100%. This high degree of neighborhood heterogeneity in both economic conditions and demographic trajectories is consistent with the understanding that complex social phenomena like gentrification, are multidimensional cultural practices rooted in the dynamics of economic restructuring, rather than simple individual choice as Zukin (1987) suggested. This observed complexity reinforces the need the need for multivariate regression modeling with appropriate controls, aimed at extracting reliable information about the underlying mechanism of the Dallas urban change.

Time-Series Charts

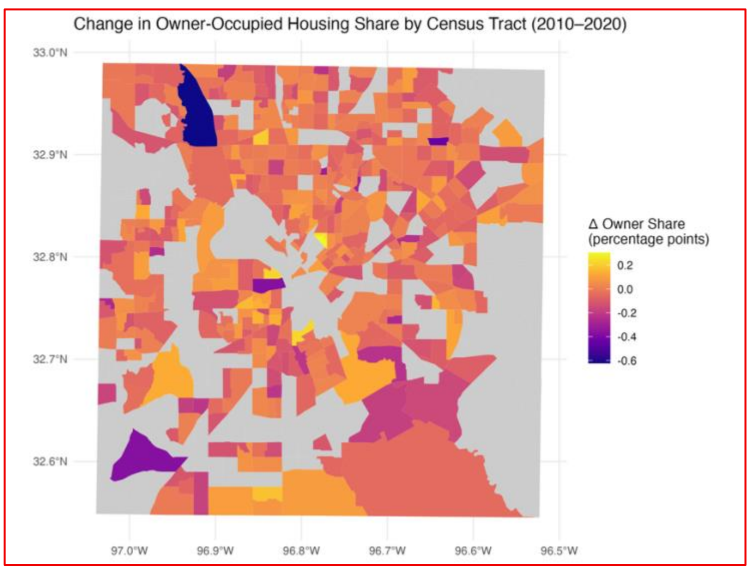


The City-Level ZHVI Trend shows the general upward trajectory of housing values in Dallas between 2000 and 2025, which confirms that housing cost pressure was a macro factor across the study period. This strong upward trend is quantitatively supported by the average log change in the ZHVI of roughly 0.88 across Dallas County tracts between 2000 and 2020, a strong indicator of long-term appreciation in housing values. This is market appreciation the broad movement of private-market investment capital into downtown districts, aligning with perspective that gentrification is a multidimensional cultural practice rooted in the dynamic economic restructuring (Zukin 1987).

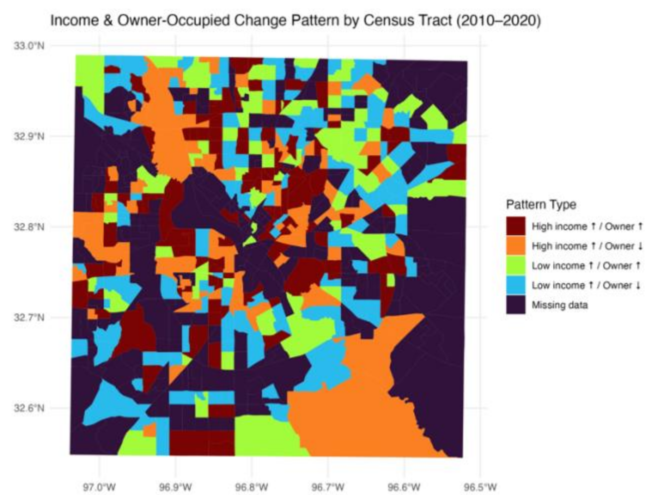


The Average Tract Income Trend demonstrated substantial growth in the average median household income across the studied Census tracts increasing from 2010 to 2020, increasing from approximately $55,000 to over $75,000 (based on the summary data). This rising income trend coupled with significant variability observed in baseline socioeconomic conditions in 2010, reinforces the necessity for multivariate regression modeling. Measuring these income shifts is crucial for understanding gentrification, a process often rooted in the expansion of the urban service economy and the private-market investment in central-city real estate (Zukin 1987).

Geospatial Maps



The Change in Owner-Occupied Housing Share highlights two distinct patterns in Dallas County between 2010 and 2020: tracts experiencing lost owner share and areas that gained it. Losses of owner share are concentrated in specific, often low-income, inner-city neighborhoods. This patterns are consistent with gentrification-like dynamics or “renter-gentrification,” where economic growth or rising income is coupled with a loss of owner-occupied housing. Such declines in homeownership, especially lower-income tracts, highlight zones of vulnerability where economic stagnation meets with reduced residential stability. Conversely, gains in owner share are clustered in the suburbs and in emerging middle-class and redeveloped urban corridors. These areas, characterized by simultaneous increases in income and owner-occupied housing, indicate stabilizing or upward mobility trends.



The derived Gentrification Pattern Types were established using a median-split approach, categorizing tracts based on whether their change in income and owner share was above or below the county median. This method effectively segments the observed diverse trajectories of socioeconomic change across Dallas County.

The tracts characterized by “High income ↑ / Owner ↑” (dark red) represent areas experiencing significant affluence and stabilization, generally clustering in suburban and emerging middle-class neighborhoods. In contrast, tracts identified by "High income ↑ / Owner ↓" (orange) are consistent with “renter-gentrification” or displacement vulnerability. In this areas, a rapidly increasing median household income is coupled with a loss of owner-occupied housing, a pattern suggesting economic restructuring where capital reinvestments occur, but the housing stock shifts towards rental units (Zukin 1987). These particular tracts often appear nearer to the urban core and typically exhibit the largest demographic shifts.

Tracts with modest income changes but increasing homeownership (green) may reflect stabilizing or transitional neighborhoods, while those identified as lower income tracts with decreasing home ownership (blue-green) highlight vulnerable zones where economic stagnation coincides with reduced residential stability. Overall, the spatial analysis reveals a complex mosaic of neighborhood change and highlights substantial neighborhood heterogeneity in both economic conditions and demographic trajectories that varies significantly across short geographic distances.

Regression Analysis (OLS)

We performed an OLS regression analysis to test whether gentrification, operationalized by rising housing values (Δ log(ZHVI)), was associated with racial/ethnic demographic change in Dallas neighborhoods between 2000 and 2020. The OLS model was specifically designed to investigate the role of housing market dynamics and baseline socioeconomic structure in shaping demographic change, testing three core relationships:

1. Whether housing price increases drive demographic change.
2. Whether baseline racial composition predicts future population shifts.
3. Whether Income and tenure structure affect who moves in/out over time.

In this model, the dependent variable is the change in Hispanic population share between 2000 and 2020 (ΔHispanic). The independent variables include: the changes in Δ log(ZHVI) between 2000 and 2020, representing long-term changes in local housing values. The baseline demographics characteristics, specifically the Hispanic percentage of the population in 2000 (Hispanic%(2000)). Baseline economic characteristics of each tract in 2010, including logged median household income (Log Income (2010)) and owner occupied housing share (Owner Share(2010)).

Variable	Estimate	Std. Error	t-value	p-value
Intercept	228.52	26.37	8.66	<0.001
Δ log(ZHVI), 2000–2020	**16.28**	7.64	2.13	**0.0345**
Hispanic % (2000)	**–0.379**	0.043	–8.83	**<0.001**
Log Income (2010)	**–22.60**	2.83	–7.98	**<0.001**
Owner Share (2010)	**36.75**	4.90	7.50	**<0.001**

The OLS regression yielded four statistically significant regarding demographic change (Δ Hispanic) between 2000 and 2020. The overall model supports the conclusion that housing market growth, initial racial composition, income levels, and homeownership structure all play important roles in shaping long-term demographic change in Dallas. The key results are as follows: (1) Housing Value Appreciation and Hispanic Growth: Changes in housing values are significantly related to changes in the Hispanic population across Dallas neighborhoods ($\beta=16.28$, $p=0.0345$). Neighborhoods experiencing faster home value growth ($\Delta\log(ZHVI)$) saw significantly larger increases in Hispanic residents. (2) Initial Demographic Composition: Neighborhoods that already had high Hispanic populations (Hispanic % (2000)) in 2000 showed smaller increases over time ($\beta=-0.379$, $p<0.001$). This outcome is generally expected because these tracts were already closer to their upper limit of population share. (3) Income and Homeownership as Predictors: Baseline socioeconomic conditions exhibited strong predictive power: Higher-income neighborhoods (Log Income (2010)) experienced declines in Hispanic population share ($\beta=-22.60$, $p<0.001$ and neighborhoods with more homeowners (Owner Share (2010)) saw large increases in Hispanic share ($\beta=36.75$, $p<0.001$). These statistically supported results indicate a more complex pattern than traditional gentrification-driven displacement narratives.

The analysis shows substantial variation among Dallas neighborhoods in both housing market change and demographic shifts from 2000 to 2020. Across all tracts, Hispanic population change ranged from a 42-percentage-point decline to a 50-point increase, while home values grew modestly on average but with considerable differences between areas (Δ log (ZHVI) mean $\approx$ 0.88). The regression results indicate that neighborhoods experiencing faster housing price growth also saw significantly larger increases in Hispanic population share ($\beta = 16.28$, $p = .035$), suggesting that appreciation in home values did not correspond to Hispanic displacement. Instead, Hispanic growth was strongest in stable, owner-occupied neighborhoods, as tracts with higher 2010 homeownership rates had substantial increases in Hispanic share ($\beta = 36.75$, $p < .001$). At the same time, high-income neighborhoods and tracts that already had large Hispanic populations in 2000 showed smaller increases—or declines—over the 20-year period.

The OLS regression findings reinforced the premises that the Dallas urban change us a multi-dimensional exhibiting substantial neighborhood heterogeneity, while showing a strong positive association between housing value appreciation ($\Delta \log(\text{ZHVI})$) and growth in population share ($\beta = 16.28$, $p = 0.035$) segments in gentrification into complex patterns rather than uniform displacement. Crucially, this growth was concentrated in stable, owner-occupied neighborhoods ($\beta=36.75, p<0.001$), suggesting that homeownership serves as a stabilizing force against displacement in areas undergoing appreciation. Conversely, tracts with higher baseline incomes showed declines in Hispanic population share ($\beta=-22.60$, $p<0.001$), aligning with spatial patterns where affluent areas continue to exclude minority in-migration. The segmented nature of change is visually confirmed by the emergence of "High income ↑ / Owner ↓" tracts near the urban core, representing “renter-gentrification” dynamics where residential stability is not achieved despite economic growth. These converged quantitative and spatial results affirm that the transformation of Dallas is channeled by local housing tenure structures, revealing intricate mechanisms that are complex, mysterious, and at least partly unknowable through simple models. Overall, the analysis moves beyond traditional displacement narratives, establishing that long-term demographic change is shaped by the interplay of economic growth, initial racial composition, income levels, and homeownership structure in a complex urban mosaic.

V. Discussion and Conclusion

Discussion of Implications

The findings from this study challenge common assumptions about gentrification and displacement in rapidly growing cities like Dallas. While rising housing values are often associated with the displacement of minority and low-income households, the results here show that increases in home values were actually linked to growth in Hispanic population share rather than decline. This pattern suggests that Hispanic households may be moving into or remaining within neighborhoods experiencing appreciation, potentially reflecting improved economic mobility, multigenerational household strategies, or the presence of affordable single-family housing stock in areas undergoing moderate reinvestment rather than full-scale luxury redevelopment. The finding that Hispanic growth is concentrated in owner-occupied neighborhoods has important policy implications. It suggests that homeownership opportunities serve as stabilizing force that enables minority households to benefit from neighborhood improvement rather than being displaced by it.

At the same time, the strong negative association between baseline income and Hispanic population growth indicates that higher-income, historically non-Hispanic neighborhoods continue to remain relatively closed to Hispanic in-migration, highlighting persistent socioeconomic barriers that limit access to more advantaged areas. The finding that Hispanic growth is concentrated in **owner-occupied neighborhoods** has important policy implications. It suggests that homeownership opportunities serve as a stabilizing force that enables minority households to benefit from neighborhood improvement rather than being displaced by it. Policies that expand access to affordable homeownership, support down-payment assistance, or protect existing homeowners from rising property taxes could therefore help sustain demographic diversity as Dallas continues to grow. Conversely, the limited Hispanic growth in high-income or high-rent areas underscores the ongoing affordability challenges faced by minority households seeking to access opportunity-rich neighborhoods. These patterns raise questions about the effectiveness of current affordable housing strategies and inclusionary zoning policies in promoting equitable neighborhood integration.

Overall, the results point to a more nuanced understanding of gentrification in Dallas. Rather than uniformly displacing Hispanic residents, neighborhood change appears to be segmented: some areas experience reinvestment that allows Hispanic households to remain or grow, while others—particularly affluent areas—continue to exclude them. This complexity suggests that Dallas' housing market transformation is not monolithic, and interventions must be tailored to the varied ways neighborhoods are changing. Future research should investigate whether similar patterns hold for other racial groups and whether these demographic changes translate into differences in long-term socioeconomic outcomes such as educational attainment, income mobility, and health. For policymakers, these findings highlight the importance of monitoring neighborhood dynamics over time and ensuring that rising property values do not create structural barriers that prevent all residents from sharing in the benefits of urban growth.

Limitations of Research

While the analysis used a reproducible and transparent pipeline, several limitations must be acknowledged. The OLS model identifies associations, but it cannot establish causation. Even if the correlation had been strong, we could not definitively state that income change caused the change in owner's share, or vice versa. The relationship could be driven by a third, unmodeled variable. The analysis of historical housing cost used MSA-level data from FRED/Zillow to provide a macro context. While necessary for a time-series perspective, this data does not capture the tract-to-tract variation in housing costs, which is a major limitation for localized gentrification studies. Future research should prioritize acquiring tract-level housing value data across the entire study period. By performing regression and descriptive statistics at the census tract level, we risk committing ecological fallacy. Tract-level averages do not necessarily reflect the experience of individuals within those tracts. For example, a rising median income could be the result of a small number of very wealthy people moving in, not a rise in income for existing residents.

Future Research

Future OLS models should incorporate the initial 2010 housing value and owner share as independent variables. These initial conditions are often powerful predictors of subsequent change and could help explain the variance that the income change variable failed to capture. The analysis used owner share as a proxy for demographic change. A more robust future study would focus on the percentage point of change of specific racial or ethnic groups to directly investigate displacement risk among minority communities, which are often most vulnerable to gentrification. Lastly, employing geographically weighted regression would allow the relationship between income change and owner share change to vary spatially. This would identify localized pockets where the relationship is strong, rather than forcing a single, weak relationship across the entire Dallas County area.

Conclusion

Our comprehensive analysis of Dallas County neighborhoods, conducted under the lens of Algorithmic Modeling Culture due to the inherent complex nature of urban nature, utilizing a hybrid strategy involving intricate record linkage to produce the primary explanatory variable the residentially weighted mean of the change in log ZHVI (Δ log (ZHVI)). Descriptive statistics confirmed substantial neighborhood heterogeneity, with overall strong long-run appreciation in real housing values averaging roughly 0.88. The OLS regression revealed a strong positive association between housing value growth and growth in Hispanic population share ($\beta=16.28, p=0.0345$), segmenting gentrification away from traditional displacement narratives. Crucially, this demographic growth was concentrated in stable, owner-occupied neighborhoods ($\beta=36.75, p<0.001$), suggesting that homeownership serves as a stabilizing force. Conversely, high-income tracts experienced declines in Hispanic share ($\beta=-22.60, p<0.001$), reinforcing persistent socioeconomic barriers and visually confirming segmented change patterns like “renter-gentrification” near the urban core. Since, neighborhoods show the results of segmented reinvestment interventions rather than monolithic policy responses. Overall, the analysis requires further research and discussion into the Dallas’s transformation about the presence of renter versus homeownership as structural stabilizer against market pressures and gentrification.

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Appendix AI Policy & Use:

In adherence to the EPPS 6302 Methods of Data Collection and Production AI Policy, this appendix will document the use of Generative AI tools during the development of the research project and throughout its data collection and production plan. AI tools have been and would be used for structural organization, rhetorical refinement, but they are not used to generate original research ideas, data generation strategies, or statistical methodology. All intellectual content is the result of human effort and academic reading.

AI Tools that have been and will be used:

- Gemini 1.5 Pro & GitHub Copilot are free for student use towards assistance on graduate-level structural organization, rhetorical and code refinement, and starter codes. Other similar AI that will be used are ChatGPT and Claude AI.

Appendix: AI-Transcripts Links

To remain fully transparent this section will include the following links to AI that were used in this project. Gemini – https://gemini.google.com/share/2b7be9ce4e40 and ChatGPT – https://chatgpt.com/share/69339bd9-3b74-8013-b10b-727ee12044e9 this was useful for our Data Analyst lead as English is her second language.

Appendix: Reproducible R/Quarto Code

Please see the following R code in the Coordinator's Quarto website: https://monicac-09.github.io/EPPS-6302/